

Mechanosensitive expression of the mesenchymal subtype marker connective tissue growth factor in glioblastoma.

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Mechanical forces created by the extracellular environment regulate biochemical signals that modulate the inter-related cellular phenotypes of morphology, proliferation, and migration. A stiff microenvironment induces glioblastoma (GBM) cells to develop prominent actin stress fibres, take on a spread morphology and adopt trapezoid shapes, when cultured in 2D, which are phenotypes characteristic of a mesenchymal cell program. The mesenchymal subtype is the most aggressive among the molecular GBM subtypes. Recurrent GBM have been reported to transition to mesenchymal. We therefore sought to test the hypothesis that stiffer microenvironments-such as those found in different brain anatomical structures and induced following treatment-contribute to the expression of markers characterising the mesenchymal subtype. We cultured primary patient-derived cell lines that reflect the three common GBM subtypes (mesenchymal, proneural and classical) on polyacrylamide (PA) hydrogels with controlled stiffnesses spanning the healthy and pathological tissue range. We then assessed the canonical mesenchymal markers Connective Tissue Growth Factor (CTGF) and yes-associated protein (YAP)/transcriptional co-activator with PDZ-binding motif (TAZ) expression, via immunofluorescence. Replating techniques and drug-mediated manipulation of the actin cytoskeleton were utilised to ascertain the response of the cells to differing mechanical environments. We demonstrate that CTGF is induced rapidly following adhesion to a rigid substrate and is independent of actin filament formation. Collectively, our data suggest that microenvironmental rigidity can stimulate expression of mesenchymal-associated molecules in GBM.

Mesenchymal transition and increased therapy resistance of glioblastoma cells is related to astrocyte reactivity.

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Grade IV astrocytoma/glioblastoma multiforme (GBM) is essentially incurable, partly due to its heterogenous nature, demonstrated even within the glioma-initiating cell (GIC) population. Increased therapy resistance of GICs is coupled to transition into a mesenchymal (MES) cell state. The GBM MES molecular signature displays a pronounced inflammatory character and its expression vary within and between tumors. Herein, we investigate how MES transition of GBM cells relates to inflammatory responses of normal astroglia. In response to CNS insults astrocytes enter a reactive cell state and participate in directing neuroinflammation and subsequent healing processes. We found that the MES signature show strong resemblance to gene programs induced in reactive astrocytes. Likewise, astrocyte reactivity gene signatures were enriched in therapy-resistant MES-like GIC clones. Variable expression of astrocyte reactivity related genes also largely defined intratumoral GBM cell heterogeneity at the single-cell level and strongly correlated with our previously defined therapy-resistance signature (based on linked molecular and functional characterization of GIC clones). In line with this, therapy-resistant MES-like GIC secreted immunoregulatory and tissue repair related proteins characteristic of astrocyte reactivity. Moreover, sensitive GIC clones could be made reactive through long-term exposure to the proinflammatory cytokine interleukin 1 beta (IL1 β). IL1 β induced a slow MES transition, increased therapy resistance, and a shift in DNA methylation profile towards that of resistant clones, which confirmed a slow reprogramming process. In summary, GICs enter through MES transition a reactive-astrocyte-like cell state, connected to therapy resistance. Thus, from a biological point of view, MES GICs would preferably

be called 'reactive GICs'. The ability of GBM cells to mimic astroglial reactivity contextualizes the immunomodulatory and microenvironment reshaping abilities of GBM cells that generate a tumor-promoting milieu. This insight will be important to guide the development of future sensitizing therapies targeting treatment-resistant relapse-driving cell populations as well as enhancing the efficiency of immunotherapies in GBM. © 2019 Pathological Society of Great Britain and Ireland. Published by John Wiley & Sons, Ltd.

Faster onset of action of formoterol versus salmeterol in patients with chronic obstructive pulmonary disease: a multicenter, randomized study.

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Chronic obstructive pulmonary disease (COPD) is a growing public health problem that has increased in recent years. It similarly affects men and women, especially those who smoke. The goals of COPD pharmacotherapy are to improve lung function, reduce symptoms, prevent exacerbations, and improve patients' health status. Bronchodilators are the foundation of treatment for COPD, and the long-acting beta2-agonists formoterol and salmeterol are both indicated for regular use by patients with stable COPD. A clinical study was conducted to compare the onset of bronchodilator effects following treatment with formoterol 12 microg administered twice-daily (BID) or salmeterol 50 microg BID. The trial also assessed whether the bronchodilator effects of treatment resulted in significant differences in clinical response. This was a randomized, multicenter, open-label, parallel-group study of formoterol 12 microg BID versus salmeterol 50 microg BID, both administered for 28 days. Patients were current or previous smokers aged ≥ 40 years, with a diagnosis of stable COPD. The primary efficacy variable was change from baseline in forced expiratory volume in 1 s (FEV1) 5 min after drug administration on day 28. Secondary efficacy variables included changes from baseline in the 6-min walk test (6MWT) and rescue medication use. The primary variable was assessed by analysis of covariance, with baseline FEV1 as the covariate. A total of 270 patients were randomized to formoterol 12 microg BID ($n=137$) or salmeterol 50 microg BID ($n=133$). In the intent-to-treat population the least square (LS) mean change from baseline in FEV1 at 5 min postdose on day 28 was 0.13 L in the formoterol group compared with 0.07 L in the salmeterol group ($P=0.022$). At 30 min postdose on day 28, the LS mean change from baseline in FEV1 was 0.17 L in the formoterol group compared with 0.07 L in the salmeterol group ($P<0.001$). Similar changes were reported at 60 min postdose (0.19 L for the formoterol group versus 0.13 L for the salmeterol group, $P=0.069$). Patients in the formoterol group walked longer distances in the 6MWT and used less rescue medication compared with patients in the salmeterol group, although the differences were not statistically significant. Significantly greater improvements from baseline in FEV1 were observed at 5 and 30 min postdose with formoterol 12 microg compared with salmeterol 50 microg after 28 days of treatment. Numeric improvements in the 6MWT and rescue medication use were also observed with formoterol.

Comparison of the effectiveness and safety of formoterol versus salmeterol in the treatment of patients with asthma: A systematic review and meta-analysis.

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Formoterol and salmeterol are two long-acting β_2 -agonists given by inhalation, with bronchodilating

effects lasting for at least 12 h after a single administration. Formoterol has a faster onset of action compared with salmeterol. The aim of this study was to perform a systematic review and meta-analysis on the data published from previous review in order to calculate pooled estimates of effectiveness and safety assessment of formoterol and salmeterol in treatment of patients with asthma. In this study, we conducted an electronic search for medical citation databases including Cochrane, PubMed, Scopus, PsycInfo, and IranMedex. Besides manual search of the databases that record randomized clinical trials, conference proceedings, and journals related to asthma were included. Studies were evaluated by two independent people based on inclusion and exclusion criteria, and the common outcomes of studies were entered into the RevMan 5.0.1 software, after evaluation of studies and extraction of data from them; and in cases where there were homogeneous studies, meta-analysis was performed, and for heterogeneous studies, the results were reported qualitatively. Of the 1539 studies initially found, 13 were included in the study. According to the meta-analysis conducted, no significant difference was found between the inhalation of formoterol 12 µg and salmeterol 50 µg in the two outcomes of mean forced expiratory volume 1 s (FEV1), 12 h after inhalation of medication and Borg score (A frequently used scale for quantifying breathlessness) after inhalation of medication. In addition, salmeterol was more effective than formoterol in the two outcomes of percent decrease in FEV1 after inhalation of methacholine and the number of days without an attack. Since the two outcomes of FEV1 30-60 min after inhalation of medication and morning peak expiratory flow after inhalation of medication were heterogeneous, they had no meta-analysis capabilities, and its results were reported qualitatively. The data from included studies shows that, more efficacy has been achieved with Salmeterol, especially in some outcomes such as the percent decrease in FEV1 after inhalation of Methacholine, and the number of days without an attack; and therefore, the administration of Salmeterol seems to be beneficial for patients, compared with Formoterol.

Faster onset of bronchodilation with formoterol than with salmeterol in patients with stable, moderate to severe COPD: results of a randomized, double-blind clinical study.

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To compare the onset and magnitude of bronchodilation after dry powder inhalations of formoterol fumarate (Foradil Aerolizer) versus salmeterol xinafoate (Serevent Diskus) with respect to normalized (*) forced expiratory volume in 1 s area under the curve 0 to 1 h after inhalation (FEV1 AUC*0-1 h). A double-blind, double-dummy, multicentre, randomized, placebo controlled, single-dose, five-period crossover study. Five centres in four countries - one centre each in France, Greece and Italy, and two centres in the Netherlands. Forty-seven patients aged 42 to 80 years (mean age 63.5 years) with chronic obstructive pulmonary disease (COPD) stage II and III, and mean baseline FEV1 1.17 L (range 0.56 to 1.77 L). Patients inhaled single doses of formoterol dry powder (12 and 24 mg), single doses of salmeterol (50 and 100 mg) and matching placebo on five separate days. The estimates of treatment difference in absolute terms (0.086 L) and percentage change from predose baseline (7.8%) for the primary end point, FEV1 AUC*0-1 h, showed that formoterol 12 mg was statistically significantly superior to salmeterol 50 mg ($P=0.0044$ and $P=0.0021$, respectively). In addition, both doses of formoterol were statistically superior to placebo for both absolute improvement and percentage change ($P=0.0001$). The analysis of secondary variables also confirmed the superiority of formoterol over salmeterol. Formoterol is associated with a faster onset of bronchodilation than salmeterol in patients with COPD.

Pharmacological similarities and differences between beta2-agonists.

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Formoterol and salmeterol are both long-acting bronchodilators that are effective in the treatment of asthma. However, some differences exist in their pharmacology that are reflected in their clinical profiles. Formoterol has a rapid onset of action, whereas salmeterol causes bronchodilation in a somewhat slower manner. However, both of these drugs are long-acting. After single doses clear effects are maintained for 12 h after inhalation, and with high doses effects are observed even at 24 h. Differences between the maximal effects of both drugs are also a consequence of their pharmacological properties. Thus, formoterol has higher intrinsic activity than salmeterol, which means that it is a full agonist, whereas salmeterol is a partial agonist on the beta2-receptor. Physicochemical properties of the drugs may explain the differences in onset and duration of action. Adequate water solubility and moderate lipophilicity of formoterol ensures rapid diffusion to the beta2-receptor on the smooth muscle and rapid bronchodilating activity. Salmeterol, on the other hand, may diffuse more slowly to the beta2-receptor because of its high lipophilicity explaining the slower onset of action. Unlike salbutamol, which is hydrophilic and has a rapid onset and short duration of action, both formoterol and salmeterol possess adequate lipophilic properties to remain in the airway tissues as a depot in close vicinity to the beta2-receptor, explaining their long duration of effect. The long duration of salmeterol has also been suggested to depend on an anchored binding within the beta2-receptor. The pharmacological evidence for a rapid onset of action of formoterol, but long duration of effect, is supported by several clinical studies. The fast onset of bronchodilation and high intrinsic activity of formoterol therefore suggest that it can be used for relief treatment in patients with asthma if they are concomitantly treated with inhaled glucocorticoids.

Why use long acting bronchodilators in chronic obstructive lung diseases? An extensive review on formoterol and salmeterol.

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Long-acting β_2 -adrenoceptor agonists, formoterol and salmeterol, represent a milestone in the treatments of chronic obstructive lung diseases. Although no specific indications concerning the choice of one molecule rather than another are provided by asthma and COPD guidelines, they present different pharmacological properties resulting in distinct clinical employment possibilities. In particular, salmeterol has a low intrinsic efficacy working as a partial receptor agonist, while formoterol is a full agonist with high intrinsic efficacy. From a clinical perspective, in the presence of low β_2 -adrenoceptors availability, like in inflamed airways, a full agonist can maintain its bronchodilatory and non-smooth muscle activities while a partial agonist may be less effective. Furthermore, formoterol presents a faster onset of action than salmeterol. This phenomenon, combined with the molecule safety profile, leads to a prompt amelioration of the symptoms, and allows using this drug in asthma as an "as needed" treatment in patients already on regular treatment. The fast onset of action and the full agonism of formoterol need to be considered in order to select the best pharmacological treatment of asthma and COPD.

[A comparison of inhalatory beta2-mimetics with long-term action (formoterol vs. salmeterol)].

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Asthma is a serious global health problem. People of all age groups in all countries of the world suffer from this chronic disease of the respiratory tract, which may have a very serious and sometimes fatal course. Most beta2-mimetics with a rapid onset of effect act for 4 to 6 hours. Inhalatory beta2-mimetics with long-term action (LABA), formoterol and salmeterol, possess a period of action longer than 12 hours. Formoterol and salmeterol represent great progress in asthma management, particularly in combination with inhalatory corticosteroids (IKS). Their effective bronchodilatory properties and long-term improvement of pulmonary functions are a great clinical contribution for the patients. Both formoterol and salmeterol are strong and effective beta2-agonists, but their different chemical structures produce different pharmacological properties. Due to the fact that the onset of the effect of salmeterol is slower, it should not be used to treat acute symptoms or quickly deteriorating asthma. Formoterol exerts a rapid onset of the effect and high internal activity, thanks to which it can be used for relieving treatment. The present paper aims to characterize and compare the properties of long-term acting beta2-mimetics, with detailed focus on their two representatives, formoterol and salmeterol.

Long acting inhaled beta-adrenoceptor agonists the comparative pharmacology of formoterol and salmeterol.

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Formoterol and salmeterol are chemically distinct, highly selective beta-2-adrenoceptor agonists developed to provide sustained (12h+) relief of airway obstruction in diseases such as asthma. Despite their similar long duration of action, these drugs differ. Formoterol has a faster onset of action in both experimental and clinical tests than that of salmeterol. Salmeterol, but not formoterol, behaves as a beta-adrenoceptor antagonist in some experimental models due to its considerably weaker efficacy at the beta 2-adrenoceptor in vitro although there are no established clinical consequences of this antagonism. Both formoterol and salmeterol display a peculiar "reassertion" behaviour in isolated airway smooth muscle subjected to beta-adrenoceptor antagonism and then washed with antagonist-free buffer. Both formoterol and salmeterol are highly efficient inhibitors of a number of indices of acute inflammatory processes in cells and tissues of human or animal origin. However, neither of these drugs has a proven clinical anti-inflammatory effect in chronic asthma in humans. Surprisingly, recent biophysical studies of formoterol and salmeterol have provided strong evidence that their individual patterns of onset speed, duration of action and "reassertion" are due to a common drug-lipid membrane interaction rather than drug-adrenoceptor interactions. A membrane-drug diffusion microkinetic model is presented to describe these phenomena.

Pharmacological basis for duration of effect: formoterol and salmeterol versus short-acting beta 2-adrenoceptor agonists.

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The mechanisms behind the long duration of bronchodilating action of the beta 2-adrenoceptor agonists formoterol and salmeterol are only partially understood. This review compares pharmacological characteristics of long-acting versus short-acting beta 2-adrenoceptor agonists in human and animal airways. Based upon the reviewed evidence, it is concluded that for beta 2-adrenoceptor agonists, long

duration of action may depend upon several factors. Both formoterol and salmeterol display a higher lipophilicity and have a higher affinity, selectivity, and potency than most short-acting agonists at the beta 2-adrenoceptor. Of these factors, lipophilicity may prove to be one of the most important ones by determining the amount of drug entering into the cell membrane in the vicinity of the beta 2-adrenoceptor. However, the receptor affinity, maximal relaxant effect (efficacy or intrinsic activity), potency, and receptor selectivity may also be of importance in determining how much beta 2-adrenoceptor agonist must remain at the receptor for sustained action.
